bits of curvature together form spiraling ridges and valleys in the fabric of spacetime.



10.1 An embedding diagram depicting the curvature of space in the orbital "plane" of a binary system made of two black holes. At the center are two pits that represent the strong spacetime curvature around the two holes. These pits are the same as encountered in previous black-hole embedding diagrams, for example, Figure 7.6. As the holes orbit each other, they create outward propagating ripples of curvature called *gravitational waves.* [Courtesy LIGO Project, California Institute of Technology.]

Since spacetime curvature is the same thing as gravity, these ripples of curvature are actually waves of gravity, or *gravitational waves.* Einstein's general theory of relativity predicts, unequivocally, that such gravitational waves must be produced whenever two black holes orbit each other—and also whenever two stars orbit each other.

As they depart for outer space, the gravitational waves push back on the holes in much the same way as a bullet kicks back on the gun that fires it. The waves' push drives the holes closer together and up to higher speeds; that is, it makes them slowly spiral inward toward each other. The inspiral gradually releases gravitational energy, with half of the released energy going into the waves and the other half into increasing the holes' orbital speeds.

The holes' inspiral is slow at first, but the closer the holes draw to each other, the faster they move, the more strongly they radiate their ripples of curvature, and the more rapidly they lose energy and spiral inward (Figures